

Data cleaning and preparation

Week Four: Digging into data

Before class begins...

- Review chapter 4 and read chapter 5 in your text.

Revisiting the data assay

Once we've found the data, figured out what is in it, and put it all together in a mineable form, we need to document its characteristics.

The data assay is the one-stop shop for variable information—the descriptive statistics, enumeration, the whole deal. Pyle adds some other information we'll talk about in a bit—but first, let's do some exploratory data mining.

Exploratory data mining (EDM)

Exploratory data mining, or EDM, is a series of summaries that attempt to discover the underlying characteristics of the process we are studying.

In general, there are two types of EDM summaries. *Parametric* summaries assume that the data belong to a family of known probability distributions, such as the normal distribution, and use measures of central tendency and variation to describe the distribution. Examples of parametric measures are the mean, median, variance, range, etc.

Non-parametric summaries look for the anchor, inflection, or cut-off points that separate the dataset into regions with equal mass—or number of data points. Examples of non-parametric summaries are histograms, data clouds, or data spheres.

When do we use these types?

The parametric approach assumes the data follow a specific functional form, and is very useful when we know something about the process that generated the data, have some previous experience with this domain, and have prior knowledge about the data structure.

The non-parametric approach does not assume any specific distribution or form. Therefore, we use this approach when we are working with unfamiliar datasets or don't know much about the data generating process.

So basically, the parametric approach is model-driven, while the non-parametric approach is data-driven.

Let's look at the parametric approach first.

Parametric approach = basic statistics

In order to understand data, we need to have a method to describe them mathematically. We call that *statistics*.

Descriptive statistics do exactly what they say – they mathematically describe the data.

Descriptive statistics can be divided into two categories:

- Measures of central tendency
- Measures of variation.

Each measure provides a piece of information about the data that is important to know.

Measures of central tendency

The most familiar measure of central tendency is the *mean*, sometimes called the *average*. The formula for calculating the mean is

$$\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i$$

What does this formula say?

- The statistical symbol for the mean is $\bar{x}$ (We call this symbol “x-bar”)
- We have N data points, and we call each data point x_i where the i takes the values $\{1, 2, 3, \text{etc.}, \text{up to } N\}$
- The Σ symbol is the Greek letter sigma, and means to add up what follows. The information at the bottom of the sigma tells where to start (in this case, at $i=1$) and the information at the top of the sigma tells where to end (in this case, at N).
- So, we start calculating x-bar by adding up the x_i from x_1 to x_N
- Then, we divide that quantity by N

Calculating the mean

Suppose we have 7 numbers:

$\{2, 7, 3, 4, 9, 6, 8\}$

(they don't need to be in any particular order)

Then, $N = 7$

$$X_1 = 2$$

$$X_2 = 7$$

$$X_3 = 3, \text{ and so on.}$$

Calculating the mean

Using the formula,

$$\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i$$

$$\bar{x} = 1/7 * (2+7+3+4+9+6+8)$$

$$\bar{x} = 1/7 * (39)$$

$$\bar{x} = 5.57$$

Measures of central tendency

The next measure is one you've probably heard about but may not be sure what it is.

We hear about the “median price” of homes – but what does that mean?

The *median* is the middle value in a group of numbers. Half the values are larger than the median, and half are smaller.

Calculating the median

There are two formulas to calculate the median, one for an odd number of values and one for an even number of values.

The values need to be ordered to calculate the median – they can't just be in any old sequence.

For odd numbers of values, the formula is:

$$\tilde{y} = y_{(N+1)/2}$$

The symbol $\tilde{y}$ is used to denote the median. Again, N is the total number of data points; so if we are finding the median price of 5 houses, $N=5$.

Calculating the median (odd)

The five houses sold for:

{ \$125,000; \$130,000; \$150,000; \$300,000; \$600,000 }

Therefore, using the formula

$$\tilde{y} = y_{(N+1)/2}$$

$$\tilde{y} = y_{(5+1)/2}$$

$$\tilde{y} = y_3$$

$$\tilde{y} = \$150,000$$

Calculating the median (even)

Suppose we had six houses that sold for:

{ \$125,000; \$130,000; \$150,000; \$300,000; \$300,000; \$600,000 }

We use the formula $\tilde{y} = \frac{y_{N/2} + y_{(N/2)+1}}{2}$

$$\tilde{y} = (y_{6/2} + y_{(6/2)+1})/2$$

$$\tilde{y} = (y_3 + y_4)/2$$

$$\tilde{y} = (\$150,000 + \$300,000)/2$$

$$\tilde{y} = \$450,000/2$$

$$\tilde{y} = \$225,000$$

What's the difference between mean and median?

Consider the housing price example:

{ \$125,000; \$130,000; \$150,000; \$300,000; \$600,000 }

- The median price is \$150,000
- The mean price is \$261,000

When there is a big difference between median and mean, there must be one or more extreme values at either or both ends of the range.

More on outliers

If you see an outlier in your data, you should investigate it. It might be “bad data” but then again, it might be *real* data.

What you do with outliers depends on your judgment of their cause and how statistically rigorous you need to be.

Can you think of examples of domains where we might find outliers that are valid values?

Measures of central tendency

The third measure of tendency is the *mode*. The mode is the value that occurs most often in a set of data points.

For example, in the second housing example,

$\{\$125,000; \$130,000; \$150,000; \$300,000; \$300,000; \$600,000\}$

the price \$300,000 appears twice. All other values only appear once. Therefore, the mode of this data is \$300,000.

We can have more than one mode in a set of data. This set of data is *bi-modal* because two values (2 and 4) appear the same number of times:

$\{2,6,5,7,2,4,9,4,0\}$

Measures of variation

These measures are also called measures of scale or spread. They show how spread out the data points are from the mean. First is the *variance*. We use the Greek lower case sigma, σ , to denote the variance. HOWEVER...

Measures of variation

In statistics, using σ implies you have data from the ENTIRE POPULATION. Most commonly, you have a SAMPLE of the population. Therefore, we use the symbol s^2 to denote the *sample variance*.*

The sample variance can be calculated using this formula:

$$s^2 = \frac{\sum_{i=1}^N (x_i - \bar{x})^2}{N-1}$$

Recall that N always stands for the total number of data points.

*Similarly, the symbol for the mean of a **population** is really the Greek letter mu (μ); $\bar{x}$ is the **sample** mean

Calculating the sample variance

Let's take a look at this formula $s^2 = \frac{\sum_{i=1}^N (x_i - \bar{x})^2}{N - 1}$

What is in the numerator? It's the sum of the squared difference between each data point and the mean, $\bar{x}$. We use $N-1$ in the denominator because we are *estimating* the mean and the variance.

Recall our first example: {2,7,3,4,9,6,8}

We determined that the mean was 5.57

So the sample variance is

$$s^2 = [(2-5.57)^2 + (7-5.57)^2 + (3-5.57)^2 + (4-5.57)^2 + (9-5.57)^2 + (6-5.57)^2 + (8-5.57)^2] / 6$$

$$s^2 = (12.76 + 2.04 + 6.61 + 11.76 + 0.18 + 2.47 + 5.9) / 6$$

$$s^2 = (41.72 / 6)$$

$$s^2 = 6.95$$

Calculating the sample variance

What happens if we don't square the difference between the mean and x_i ?

$$s^2 = [(2-5.57) + (7-5.57) + (3-5.57) + (4-5.57) + (9-5.57) + (6-5.57) + (8-5.57)]/6$$

$$s^2 = (-3.57 + 1.43 + (-2.57) + (-1.57) + 3.43 + 0.43 + 2.43)/6$$

$$s^2 = (0/6)$$

$$s^2 = 0$$

We know that can't be the right answer!

Measures of variability

The second common measure of spread is the standard deviation.

Once you've calculated the variance, finding the standard deviation is easy – it's just the square root:

$$s = \sqrt{s^2} \text{ for the sample standard deviation}$$

$$\sigma = \sqrt{\sigma^2} \text{ for the population standard deviation.}$$

For our example, {2,7,3,4,9,6,8}, the standard deviation is

$$s = \sqrt{6.95}$$

$$s = 2.64$$

The significance of the standard deviation

In a normal, or bell-shaped distribution of data, about 68% of the data will fall within 1 standard deviation, or sigma, of the mean.

99.7% will fall within 3 sigmas of the mean (6 sigmas total).

Fraction of data	Number of Standard Deviations from Mean
50.0%	0.674
68.3%	1.000
90.0%	1.645
95.0%	1.960
95.4%	2.000
98.0%	2.326
99.0%	2.576
99.7%	3.000

The significance of the standard deviation

Even for our little 7-value example, this fact still holds

$$\{2, 7, 3, 4, 9, 6, 8\}$$

Mean=5.57

Standard deviation = 2.64

68% of 7, or ~5 values, should be within 1 standard deviation of the mean

$$5.57 \pm 2.64 = \{2.93, 8.21\}$$

3, 4, 6, 7, and 8 fall in that range.

Other measures of variation

We know how to get the mean, median, variance, and range (just max-min). Some other parameters that are useful are quartiles and the inter-quartile range.

There are several ways to determine the quartiles (also called percentages). An easy way is to take the median of the lower half of the data for the lower quartile Q_1 , then take the median of the upper half of the data for the upper quartile Q_3 .

Finding the quartiles

Let's use this dataset:

{85 69 69 74 51 85 81 96 64 84 77 78 117 73 73 85 87 84 95 84}

Remember, to find the median we need to order the data.

{51 64 69 69 73 73 74 77 78 81 84 84 84 85 85 85 87 95 96 117}

The median for this data is 82.5.

{**51 64 69 69 73 73 74 77 78 81** **84 84 84 85 85 85 87 95 96 117**}

The green numbers are to the left of the median, and we use those to find the lower quartile. For these data, the lower quartile Q1 is **73**.

The blue numbers are to the right of the median, and we use those to find the upper quartile. For these data, the upper quartile Q3 is **85**.

See *quartiles.xls* for different ways of calculating the quartiles.

Interquartile range

The interquartile range, or IQR, is a more robust way of determining the range of the data because it ignores outliers. The IQR is simply the difference between the third quartile and the first.

Suppose we have this dataset: {1, 2, 2, 3, 4, 6, 6, 80}* The range is $80 - 1 = 79$; we can see that the 80 has an inordinate effect on this parameter. Let's look at the IQR. The first quartile, Q1, is 2. The third, Q3, is 6. So,

$$= 6 - 2$$

$$= 4$$

$$\widehat{\text{IQR}} = Q3 - Q1$$

Why do we have a "hat" on IQR?

*Dasu and Johnson, pp. 40-41.

Estimates and confidence

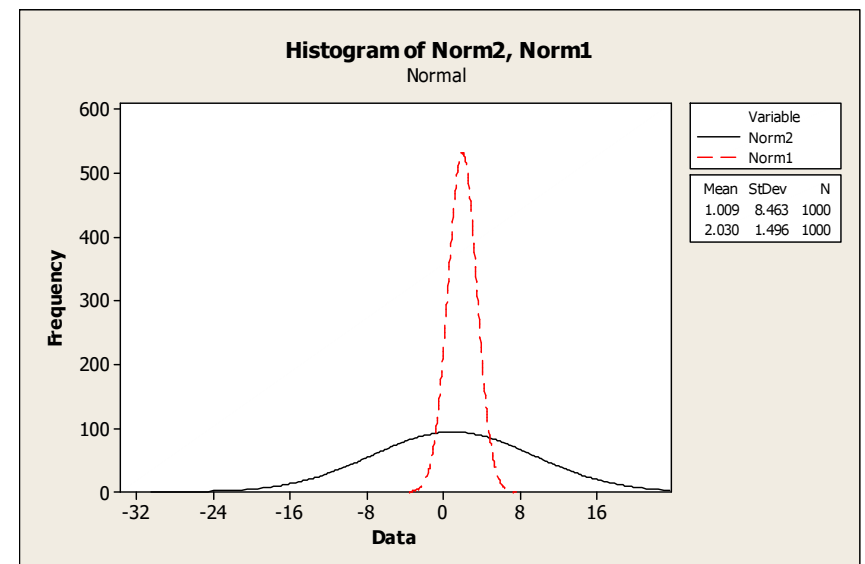
These parameters are *point estimates* that give us an idea how the data are distributed. But as estimates, they have an inherent fuzziness (recall your pulse measurements from last week). We want to know how fuzzy they are, and we also want to know if estimates from one sample of data match the estimates of another.

In order to find out, we have to know something about the normal distribution and what confidence means, statistically speaking.

Introducing the normal distribution

In real life situations, many continuous random variables are distributed normally. The relative frequency plot of these variables forms a bell-shaped (or Gaussian) curve. Some examples would be heights of people, IQ scores, birth weights, or cholesterol levels.

This graph shows two normal distributions with different means and standard deviations, which affects the shape of the bell curve.



Properties of the normal distribution

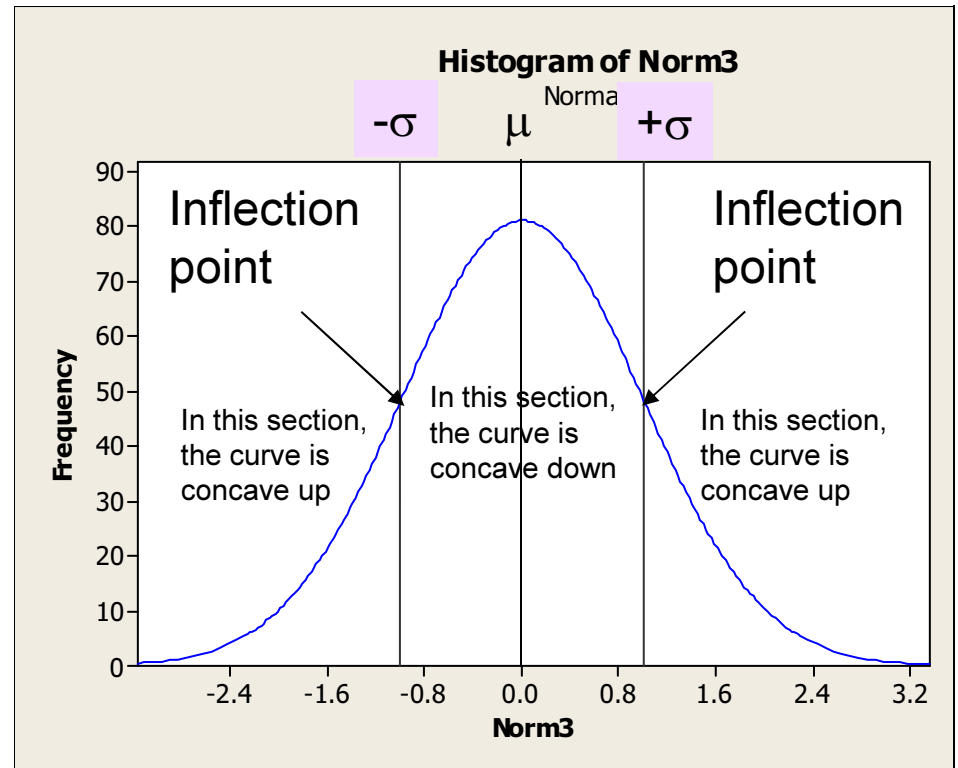
Every normal distribution exhibits specific characteristics.

First, the curve is symmetric around the mean.

Second, the mean is the highest point in the curve.

Third, the curve has inflection points at $(\mu + \sigma)$ and $(\mu - \sigma)$.

That means the curve changes direction at those points.



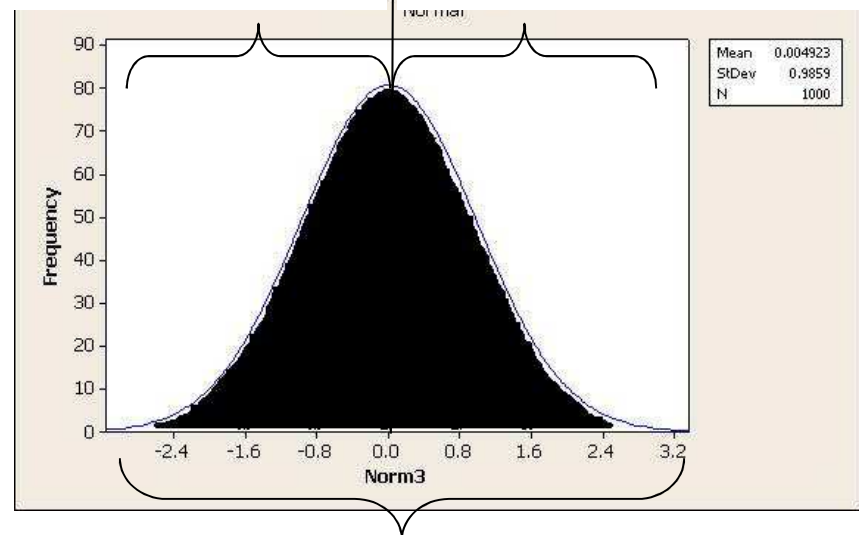
Properties of the normal distribution

Fourth, the area under the curve equals 1.

Fifth, because the curve is symmetric, the area to the right of μ = the area to the left of μ = 0.5

Sixth, as x increases (or decreases) toward infinity, the curve approaches *but never reaches* $y=0$.

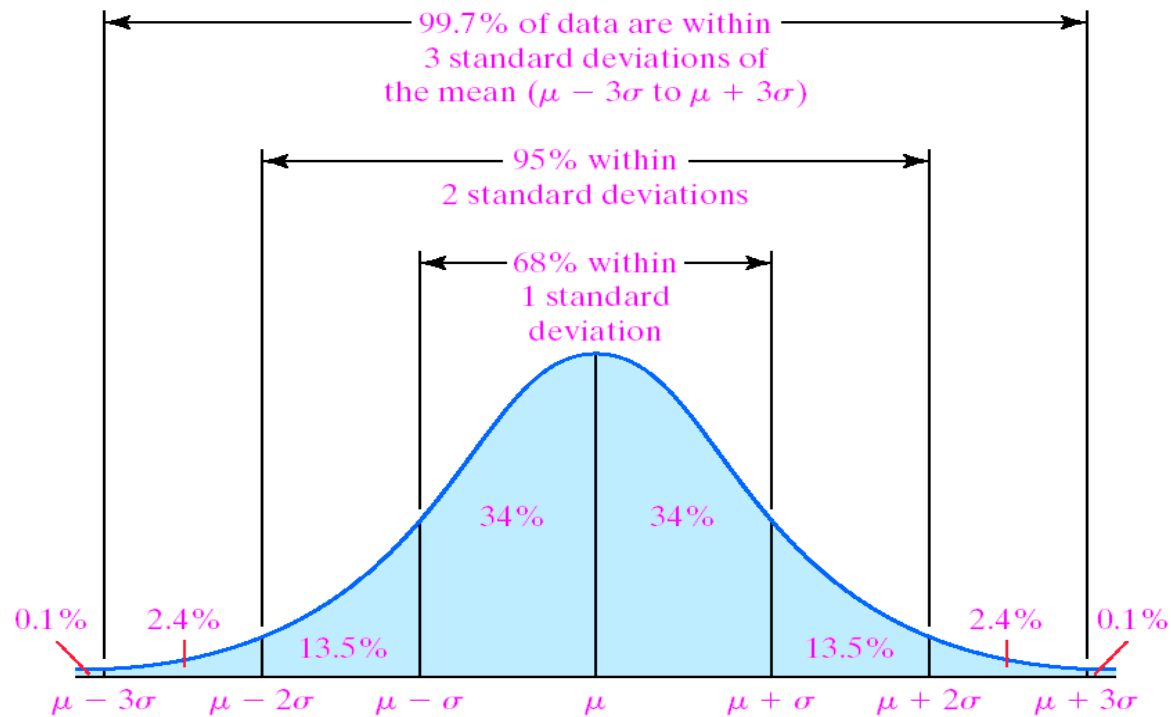
Area under curve = 0.5 Area under curve = 0.5



Area under curve = 1

More on the normal distribution

Note that the Empirical Rule applies to the normal distribution.



The area under the curve

If a random variable X is normally distributed, with mean μ and standard deviation σ , then the area under the curve for given values of X represents the percentage of the population (or the probability of an individual observation) that will fall in that range.

This concept is very important to understand. It allows us to ask questions about the data and the distribution, and to make inferences about the population.

The area under the curve

For example, we might be interested in finding out:

- What is the probability that the height of 2 year old is greater than 37”?
- Or, what proportion of the 2-year-old child population is between 33 and 36 inches?
- Or what is the probability that an individual child will be less than 35” tall?

In order to answer these questions and make use of the properties of the normal distribution, we need to **standardize** the random variable X (heights) using the z-score.

The z-score

The **z-score** represents the number of standard deviations that a data value is from the mean. We find the z-score by subtracting the mean from the individual data values and dividing this result by the standard deviation.

Either formula can be used depending upon whether we are using population data or sample data.

Population data

$$z = \frac{x - \mu}{\sigma}$$

Sample data

$$z = \frac{x - \bar{x}}{s}$$

Remember:

x is the individual data point

μ is the population mean

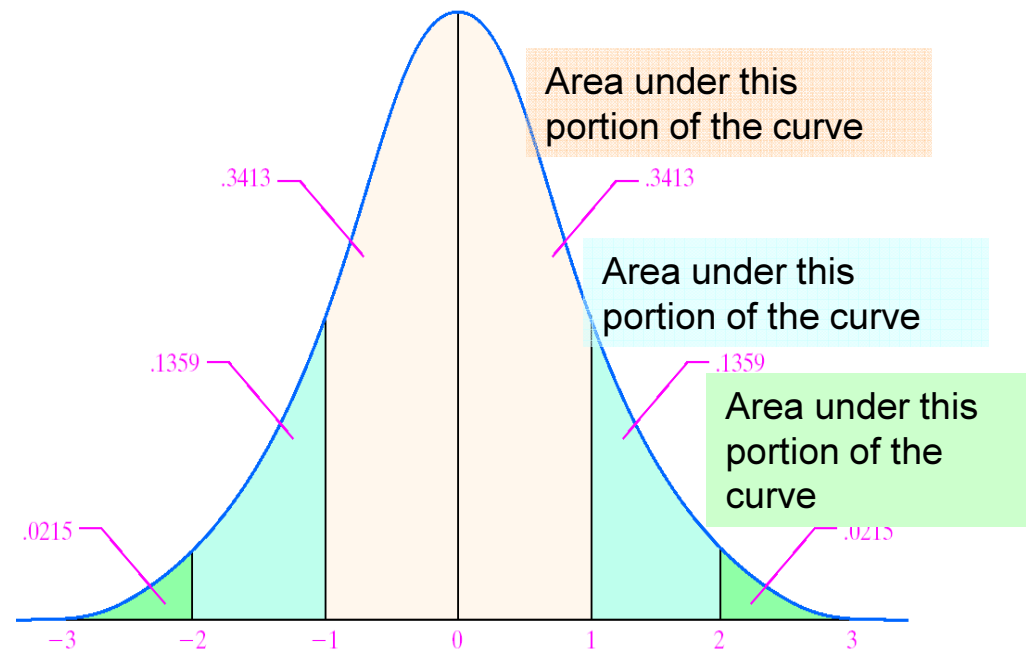
σ is the population standard deviation

x bar is the sample mean

s is the sample standard deviation

Using the z-score

The distribution generated using the z-score is called the “standard normal”; it has a mean of 0 and a standard deviation of 1. We can apply the Empirical Rule to this distribution as well. The x-axis in this figure is the distance from the mean in units of standard deviation. These distances are also the z-scores.



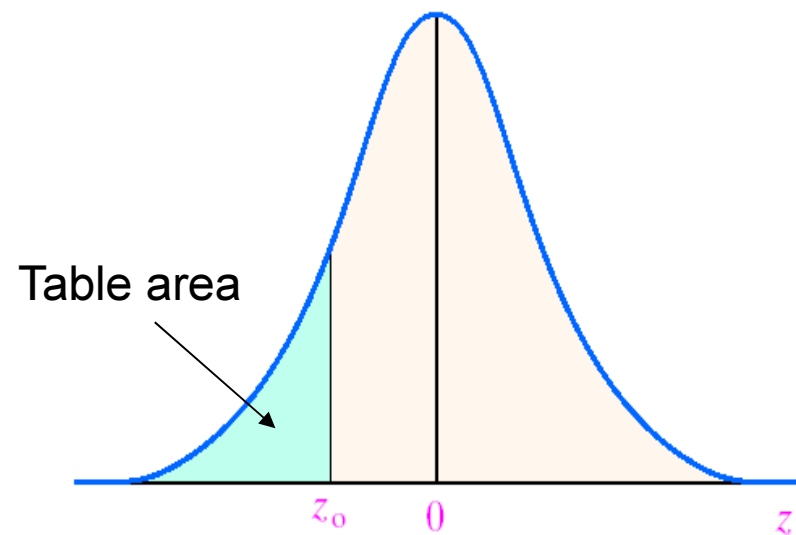
Looking up z-scores

We use standard tables to find the areas under the curve that correspond to specific z-scores. Every statistics book has these tables; some have a full table that covers from $Z = -3.00$ to $+3.00$. To save space, and because the area under the curve is symmetric, some books have a half table that covers from $Z = 0$ to $+3.00$.

How to use the tables

A z-table gives the area under the standard normal curve for values to the *left* of a specified z-score, z_0 , as shown in the figure. The table area is the light green area.

Note that negative z-scores correspond to areas less than 0.5, while positive z-scores correspond to areas greater than 0.5



Reading the table

To find the area, look for the first two digits of the z-score in the leftmost column. Then follow across that row to the column corresponding to the third digit.

For example,

$$z(1.00) = 0.8413$$

$$z(0.46) = 0.6772$$

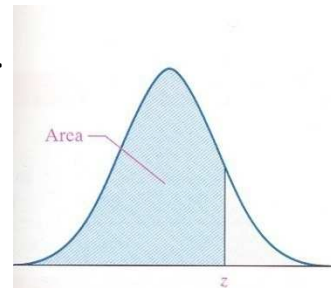


Table IV (continued)

Standard Normal Distribution

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319

Reading the table

$$z(-2.95) = 0.0016$$

$$z(-3.12) = 0.0009$$

What is $z(-2.17)$?

$z(-3.44)$?

Remember, these values are the area to the *left* of the z-score.

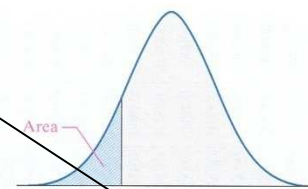


Table IV

Standard Normal Distribution

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
-3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
-3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
-3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
-3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
-2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
-2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
-2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
-2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
-2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
-2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
-2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
-2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
-2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
-2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183

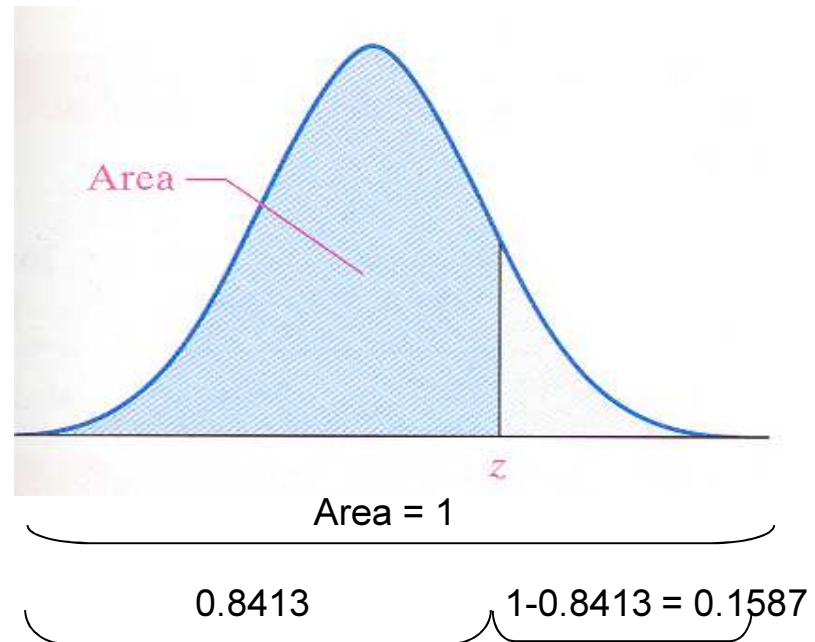
Which of these three questions could be answered using this formula?

1. What is the probability that the height of 2 year old is greater than 37"?
2. What proportion of the 2-year-old child population is between 33 and 36 inches?
3. What is the probability that an individual child will be less than 35" tall?

What if we want to find the area to the *right* of the z-score?

In that case, we would find the area to the left of the z-score and subtract it from 1. That's because the total area under the curve is 1.

Suppose z_0 is 1.00; the table shows the area to the left of $z_0 = 0.8413$. Therefore, according to the property of standard normal distributions that the area under the curve equals 1.00, the area to the right is $1 - 0.8413 = 0.1587$

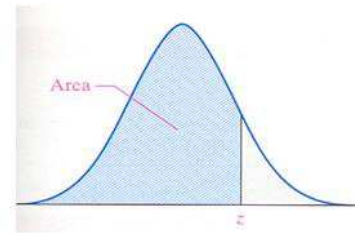


Which of these three questions could be answered using this formula?

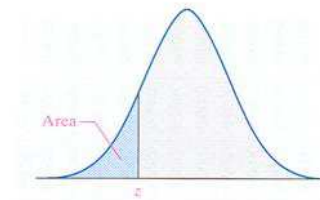
1. What is the probability that the height of 2 year old is greater than 37"?
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Finding the area between two z-scores

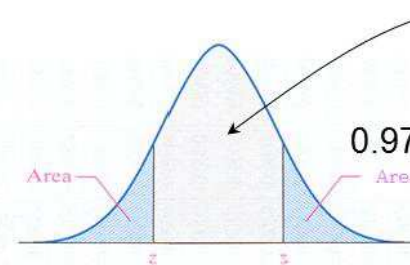
- Start with the rightmost Z score and find the area under the curve to the left of that score.
- Then find the area under the curve to the left of the leftmost Z score.
- Subtract the two areas – this result gives you the area in the middle.



$$Z(2.01) = 0.9778$$



$$Z(-1.35) = 0.0885$$



$$0.9778 - 0.0885 = 0.8893$$

Which of these three questions could be answered using this formula?

1. What is the probability that the height of 2 year old is greater than 37"?
2. What proportion of the 2-year-old child population is between 33 and 36 inches?
3. What is the probability that an individual child will be less than 35" tall?

Finding the z-score from the area

Sometimes we want to go the other direction—find the z-score, given the area. In that case, find the closest number to the given area on the table and note the row and column labels. Those labels give you the corresponding z-score.

For example, suppose you want to find the z-score for the area 0.9220. The closest value on the table is 0.9222, which corresponds to a z-score of 1.42

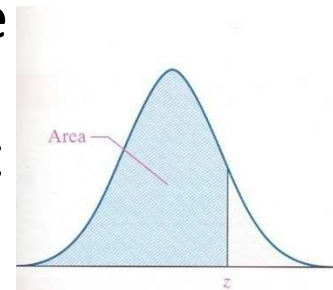


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0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9333	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441

Interpreting the area under the curve

The area under the standard normal curve can be interpreted as a probability or the proportion of the population. For instance, we can say “the percent of the population with a z-score of at most 1.0 is 84%.” Or we can say “the probability associated with an individual having a z-score of at most 1.0 is 0.84”.

Confidence intervals around the mean

We have talked about the population parameters μ and σ . We've learned that we can generate a random sample of a population and estimate μ and σ using $\bar{x}$ and s .

$\bar{x}$ and s are called *point estimators* for μ and σ . We also say they are *unbiased* point estimators, in that they do not over- or under-estimate the value of the parameters. That can be a confusing statement, but what it essentially means is that the expected value for $\bar{x}$ and s would yield the correct μ and σ . The differences we see between the estimates and the parameters can be explained by variability in the random samples we use to calculate the estimates, not in the formulas for the estimates.

The effects of sampling variability

To show how the choice of instances in random sampling affects the estimates, here are two sets of test grades for my data mining class. The grades were not sorted and I chose every fifth grade. For Sample 1, I started with the first grade in the list. For Sample 2, I started with the fifth grade in the list.

	Sample 1	Sample 2
	60.5	99
	32	83.5
	82.5	72
	78	83
	82	78
Average	67	83.1
Std Dev	19.25	10.03

Class average	78.60
Class st dev	13.39

Overcoming the effects of sampling variability

My class had 30 students in it, so calculating the true class μ and σ is not burdensome (unlike actually grading the tests). But what if I wanted to know the μ and σ of all students taking introductory data mining classes? I couldn't say with certainty that all such classes would have the same parameter values as mine does, but I could use my class as the sample, and develop a *confidence interval* about the mean that would represent the probability that the true mean lies somewhere in a range above and below my mean.

So we would say something like, “*I am 95% sure that the true mean lies somewhere in the interval between 75 and 80*”. That statement also says I am 95% sure that there's a 5% chance I am wrong.

More about confidence intervals

So, confidence intervals have levels of confidence (or probabilities) associated with them. Usually these levels are 90%, 95% or 99%. We express these levels statistically as $(1 - \alpha)$ {One minus alpha} levels where α is the chance you are wrong. A 95% confidence interval (CI) means there is a 5% chance you are wrong. A 90% confidence interval (CI) means there is a 10% chance you are wrong.

There are two ways to interpret confidence intervals (in order of importance):

1. A 95% confidence interval means I am 95% certain that the interval contains the true population mean μ .
2. If 100 confidence intervals were constructed from 100 different samples, we expect 95 of 100 intervals to contain the true population mean μ . The other 5 intervals will not contain μ .

Constructing confidence intervals

First, we will look at the case where we know σ (Note that in real life situations we rarely do know what σ is).

We need two other parameters to find the confidence intervals: $\bar{x}$, or the sample mean, and α , our confidence level.

The confidence interval is always centered around $\bar{x}$. Because the interval is symmetric around $\bar{x}$, we divide α by 2 and find the upper and lower limits by using the z-score for our chosen $\alpha/2$.

The equation for the upper interval boundary is $\bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$

Similarly, the equation for the lower interval boundary is $\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$

We often use shorthand for these equations and state them as

$$\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

An example

Let's construct a 95% confidence interval for my class exam data. We know that $\sigma = 13.39$. Using Sample 2's parameters, $\bar{x} = 83.1$ and $n = 5$.

The upper confidence interval boundary is

$$\begin{aligned}\bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \\ = 83.1 + 1.96 \frac{13.39}{\sqrt{5}} \\ = 94.84\end{aligned}$$

The lower confidence interval boundary is

$$\begin{aligned}\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \\ = 83.1 - 1.96 \frac{13.39}{\sqrt{5}} \\ = 71.36\end{aligned}$$

Therefore, we say that the 95% confidence interval for the exam scores is (71.36, 94.84).

What happens if we use Sample 1's $\bar{x}$? The 95% confidence interval is (55.26, 78.74). In both cases, the interval covers the true mean.

What happens as confidence decreases?

Suppose we were less certain and wanted a 90% confidence interval. What would the interval be for Sample 1 data?

Calculate the interval (hint: the z-score is 1.65) and see what happens. Then try with 99% confidence ($z = 2.58$). Can you explain what happens as confidence levels change?

Finding confidence intervals when σ is unknown

If we don't know σ , then we have to use s , the sample standard deviation. That means we don't use the z statistic any more; we use the t statistic, which is based on Student's t distribution.

William Gosset

Born: 1876 in Canterbury; **Died:** 1937 in Beaconsfield, England

Obtained a post as a chemist in the Guinness brewery in Dublin in 1899. Invented the t -test to handle small samples for quality control in brewing. Wrote under the name "Student" (pronounced "ShtuDENT", like the German pronunciation).



Student's t distribution

The t distribution has many characteristics that are similar to the normal distribution. And actually, it looks a lot like the normal but with thicker tails.

Properties of the t distribution:

- The mean = 0
- It is symmetric about the mean
- It forms a family of distributions for each degree of freedom (degrees of freedom (df) = $n-1$).
- As df increases, the distribution looks more normal
- Its standard deviation σ is >1 , but as df increases σ gets closer to 1.0, like the standard normal
- It peaks at the mean like the normal but is thicker at the tails because $\sigma > 1$.

What are degrees of freedom?

Degrees of freedom refers to the number of unconstrained values (I know, that sounds confusing—just bear with me for a moment). The idea is that when we find the mean of a set of values, we then can calculate the residuals, or the difference between each value and the mean. Mathematically, the residuals have to sum to zero. Suppose we have 4 values: 2, 4, 5, and 9. The mean is 5. The residuals for this dataset, then, are

$$2-5 = -3 \text{ (sum of residuals} = -3)$$

$$4-5 = -1 \text{ (sum of residuals} = -3 + -1 = -4)$$

$$5-5 = 0 \text{ (sum of residuals} = -3 + -1 + 0 = -4)...$$

Stop right there! We don't have to calculate the residual for the last value, 9, because we already know what it has to be (that is, it's constrained) in order for the sum of the residuals to equal zero. Therefore, this four-number dataset has three unconstrained values—or, three degrees of freedom.

Constructing the confidence interval

To construct a confidence interval with σ unknown we use the equation:

$$\bar{x} \pm t_{df, \alpha/2} \frac{s}{\sqrt{n}}$$

In order to use this equation you need to be able to look up the values of $t_{df, \alpha/2}$ in a t distribution table (you can find one [here](#)). The t values give you the area under the curve, just like the z scores do, so the values are symmetric and we use them the same way we used the z score. Read the degrees of freedom down the first column on the left of the table, and the area (which corresponds to $\alpha/2$) across the top.

Example

Let's use the original Sample 2 test data again. The n was 5, so our $df = n-1 = 4$. Look up the t value for $df = 4$ and $\alpha = 0.5$. Recall that we are looking for the area under the curve between $\alpha/2$ and $1 - \alpha/2$, just like the z -score example.

The $t_{4, 0.25} = 2.776$, so the confidence interval is

$$\begin{aligned} &= \bar{x} - t_{df, \alpha/2} \frac{s}{\sqrt{n}} \\ &= 83.1 - 2.776 \frac{10.03}{\sqrt{5}} \\ &= 70.65 \end{aligned}$$

$$\begin{aligned} &= \bar{x} + t_{df, \alpha/2} \frac{s}{\sqrt{n}} \\ &= 83.1 + 2.776 \frac{10.03}{\sqrt{5}} \\ &= 95.55 \end{aligned}$$

	Sample 1	Sample 2
	60.5	99
	32	83.5
	82.5	72
	78	83
	82	78
Average	67	83.1
Std Dev	19.25	10.03

Confidence intervals around the variance

Instead of the standard normal distribution, we must use the chi squared distribution to find confidence intervals around the variance and standard deviation.

The procedure is generally the same as finding the confidence interval around the mean, except for two things:

1. The data must be normally distributed, or the confidence interval makes no sense, and
2. Since the chi-square distribution is not symmetric, we have to calculate lower and upper limits separately.

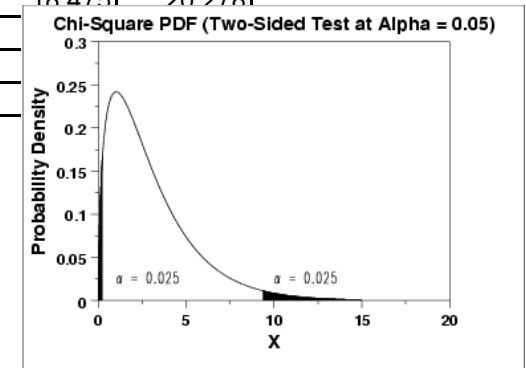
Confidence intervals around the variance

To compute the lower limit, we use the expression $LL = \frac{(n-1)s^2}{X^2_{\alpha/2, n-1}}$

The upper limit is $UL = \frac{(n-1)s^2}{X^2_{1-\alpha/2, n-1}}$

Just as in the case of the t distribution, we find the X^2 values from a [table](#).

The areas given across the top are the areas to the right of the critical value. To look up an area on the left, subtract it from one, and then look it up (ie: 0.05 on the left is 0.95 on the right)										
df	0.995	0.99	0.975	0.95	0.9	0.1	0.05	0.025	0.01	0.005
1	---	---	0.001	0.004	0.016	2.706	3.841	5.024	6.635	7.879
2	0.01	0.02	0.051	0.103	0.211	4.605	5.991	7.378	9.21	10.597
3	0.072	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345	12.838
4	0.207	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277	14.86
5	0.412	0.554	0.831	1.145	1.61	9.236	11.07	12.833	15.086	16.75
6	0.676	0.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812	18.548
7	0.989	1.239	1.69	2.167	2.833	12.017	14.067	16.013	18.475	20.278
8	1.344	1.646	2.18	2.733	3.49	13.362	15.507	17.535		
9	1.735	2.088	2.7	3.325	4.168	14.684	16.919	19.023		
10	2.156	2.558	3.247	3.94	4.865	15.987	18.307	20.483		



Caveat about confidence intervals and variance

For the confidence interval around the variance, the data must be normally distributed. If they are not, you will have to do some transformations (which is beyond the scope of this course).

You can use the formula in the text to determine the minimum number of instances to reach a specific confidence level for an attribute.

Sampling and variability

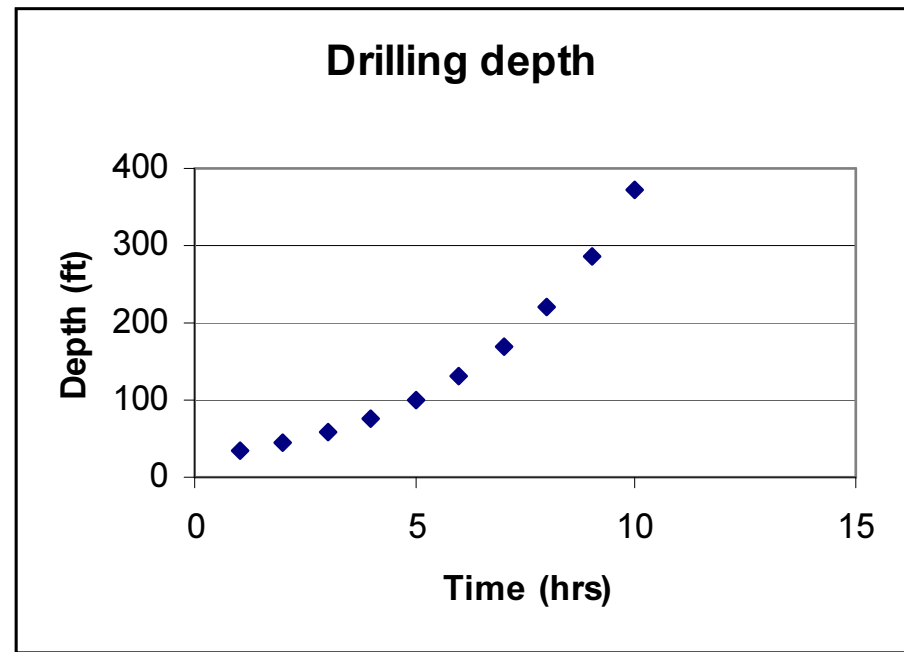
The author sort of dances around the idea of the Law of Large Numbers—as the number of instances increases, the descriptive statistics converge to a particular value (with a certain amount of error). The trick is to make sure we have captured the variability in the data (the stuff that makes it interesting) and that there is sufficient variability to show that patterns do exist in the dataset.

Relationships between two variables

Many times we are interested in finding out how one factor influences another. For instance, we might want to see how drilling depth changes over time. That means we have two variables, time and depth, that are related to each other. We call this *bivariate data* and express it as an ordered pair (x, y) . The convention is that x is the *independent*, or *predictor*, variable; y is the *dependent*, or *response* variable. We want to see how the change in x affects the value of y .

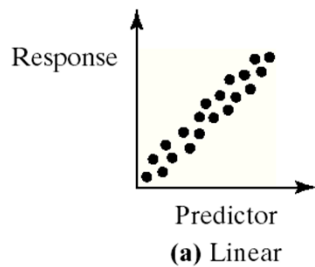
Scatter plots

Scatter plots are often used to graphically represent this paired data. We plot the independent variable, x , on the horizontal axis and the dependent variable, y , on the vertical axis.

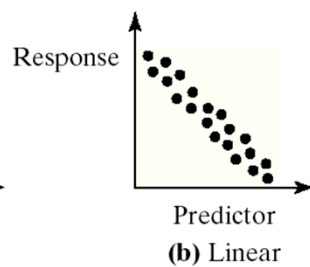


Showing relationship

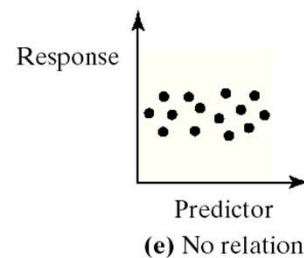
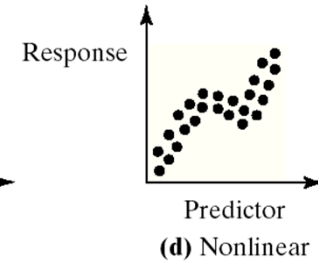
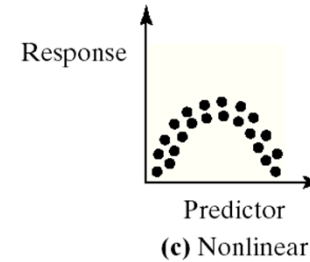
These scatter plots show different relationships that can exist between two variables



Positive correlation
(going up)



Negative correlation
(going down)



Correlations

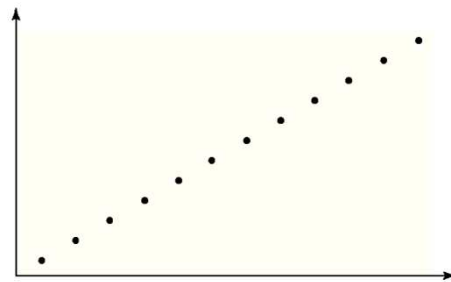
Two variables are *positively correlated* if an increase in x triggers an increase in y . They are *negatively correlated* if an increase in x results in a decrease in y . The *linear correlation coefficient* is a measure of the strength of linear relation between two quantitative variables. Here is the formula for the sample correlation coefficient.

$$r = \frac{\sum \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{y_i - \bar{y}}{s_y} \right)}{n-1}$$

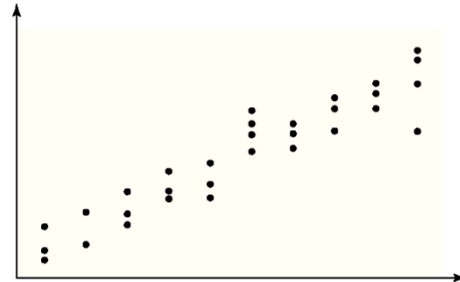
Some rules about “ r ”

1. The linear correlation coefficient is always between -1 and 1, inclusive. That is, $-1 \leq r \leq 1$.
2. If $r = +1$, there is a perfect positive linear relation between the two variables.
3. If $r = -1$, there is a perfect negative linear relation between the two variables.
4. The closer r is to +1, the stronger the evidence of positive association between the two variables.
5. The closer r is to -1, the stronger the evidence of negative association between the two variables.
6. If r is close to 0, there is evidence of no linear relation between the two variables. Because the linear correlation coefficient is a measure of strength of linear relation, r close to 0 does not imply no relation, just no linear relation.

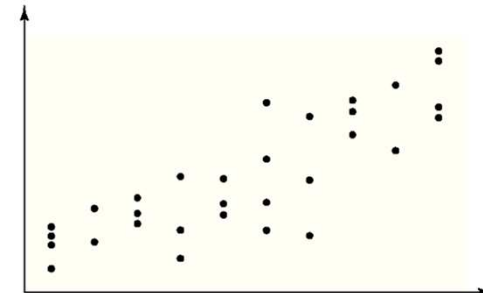
Examples of correlation



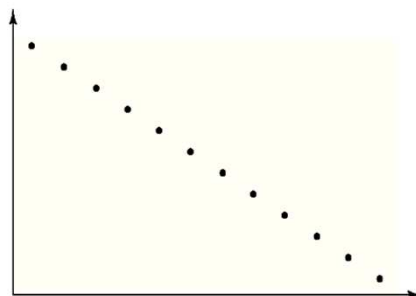
(a) Perfect positive linear correlation, $r = 1$



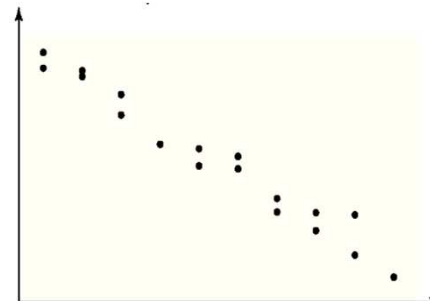
(b) Strong positive linear correlation, $r \approx 0.9$



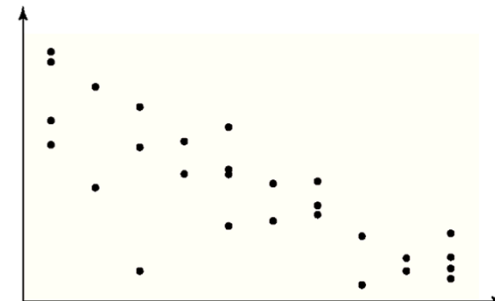
(c) Weak positive linear correlation, $r \approx 0.4$



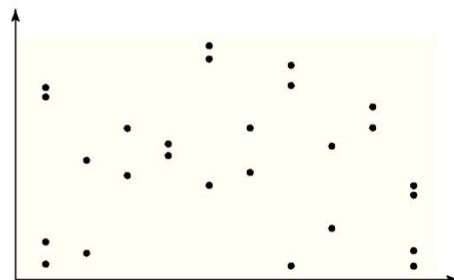
(d) Perfect negative linear correlation, $r = -1$



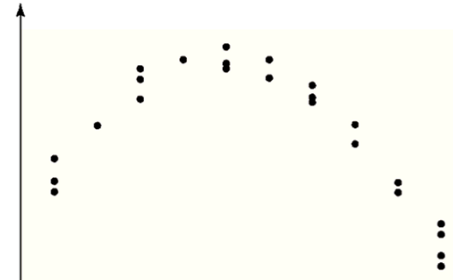
(e) Strong negative linear correlation, $r \approx -0.9$



(f) Weak negative linear correlation, $r \approx -0.4$



(g) No linear correlation, r close to 0.



(h) No linear correlation, r close to 0.

Correlation in Minitab

Minitab calls the correlation coefficient the *Pearson correlation*. To compute it, choose

Stat > Basic stat > Correlation

In real life a correlation coefficient of approximately (+ or –) 0.7 to (+ or -)1.0 is pretty well correlated. A coefficient does not have to be 0.9 to say the correlation is good. Text books will have problems that show that kind of correlation. Real life is different.

Correlation and causation

Be careful not to confuse *correlation* (association between the values of two variables) with *causation* (x making y happen). They are not always the same thing. For instance, in the summer there is an increase in drowning deaths. There is also an increase in ice cream sales. The two are related (by the common time variable) but we wouldn't say one causes the other.

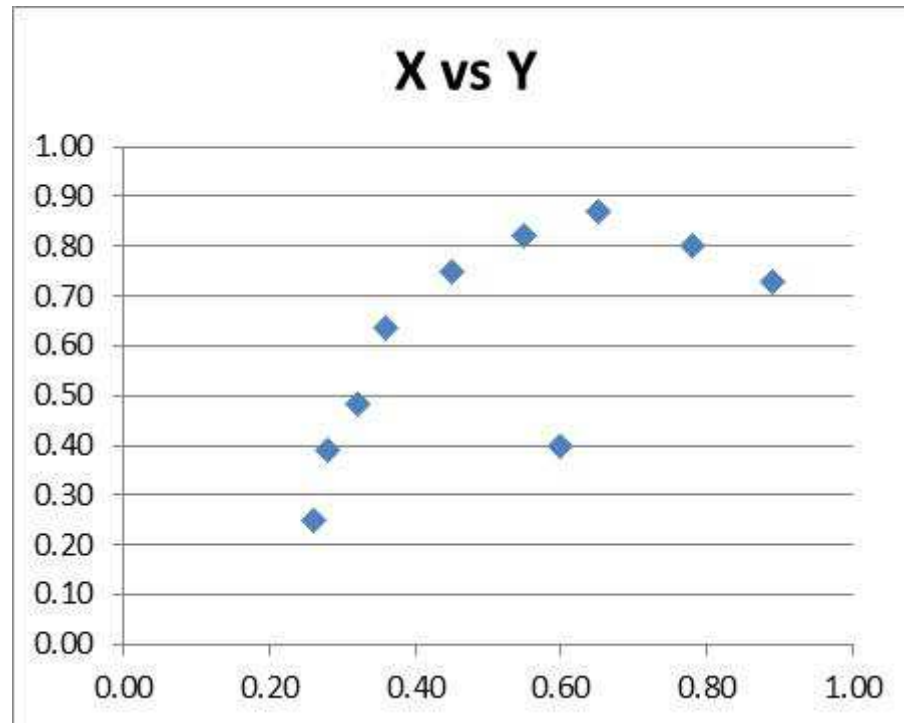
Scatterplots and finding outliers

When we are exploring our data we should look at our data graphically as well as numerically. Plotting helps us see patterns and anomalies in the data that we would not otherwise notice—especially when we plot two attributes against each other. Here is a small dataset of two attributes. Both have roughly the same range, mean, and standard deviation. No outliers here, right?

	X	Y
	0.45	0.75
	0.78	0.80
	0.26	0.25
	0.65	0.87
	0.36	0.63
	0.32	0.48
	0.60	0.40
	0.55	0.82
	0.28	0.39
	0.89	0.73
Min	0.26	0.25
Max	0.89	0.87
Mean	0.51	0.61
St Dev	0.217	0.216

Scatterplots and finding outliers

If you look at each attribute individually, you won't see this relationship, or this outlier.



Graphing data

Weka will generate a bivariate plot, as will several other software packages. Be careful though, to make sure the graph is correct. Watch those axes!

See the optional lecture material and exercise on graphing data in this week's content section.

Back to the data assay

What was all that statistics material for? In the data assay we use these concepts to determine the characteristics of variables. If you look at the table on page 143 you can see the different statistics used in the data assay for the CREDIT data file.

The data assay is your dictionary for a particular dataset. It will help you figure out what to do about defects in the dataset, such as those pesky missing values and outliers.

Summary

Chapters 4 and 5 introduce the basic data exploration that is necessary to help us understand what the dataset contains, and to determine if the dataset is interesting enough to continue with the mining process.

In the next lesson we'll look at ways to characterize non-numeric data.

Discussions for this week

Discussion question: Time to get your hands dirty!

Download the dataset Diabetes.xls from MyCourses. This dataset looks at the incidence of diabetes in Pima Indian women. The class variable is binary (whether or not the individual has diabetes).

Choose one of the attributes and report its descriptive statistics. Then choose another attribute and compare the two together. Discuss your findings and observations about this dataset.

Homework 2: Due 2/24

Using the dataset from HW 1, expand your data dictionary. For each attribute,

- For all continuous attributes, calculate
 - Measures of central tendency
 - Measures of variation
 - Quartiles and inter-quartile range
 - Confidence intervals about the mean
 - Generate bivariate attribute plots and note any interesting findings.
- Write up your results as if you were reporting to your manager. Be thorough in your evaluation of the dataset.
- Submit your data dictionary to the HW2 dropbox.